

Contents

1	Continuous Functions	2
1.1	Continuity of a Function	2
1.2	Homeomorphisms	3
1.3	Constructing Continuous Functions	3

Continuous Functions

1.1 Continuity of a Function

Definition 1.1.1

Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is said to be **continuous** if for each open subset V of Y , the set $f^{-1}(V)$ is an open subset of X .

Theorem 1.1.2

Let X and Y be topological spaces; let $f : X \rightarrow Y$. Then the following are equivalent:

1. f is continuous.
2. For every subset A of X , one has $f(\bar{A}) \subset \overline{f(A)}$.
3. For every closed set B of Y , the set $f^{-1}(B)$ is closed in X .
4. For each $x \in X$ and each neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subset V$.

Remark.

If the condition in (4) holds for the point x of X , we say that f is **continuous at the point x** .

Proof. 1. $\implies$ 2. : If f is continuous, take any $x \in f(\bar{A})$. Let V be any neighborhood of $f(x)$. Then $f^{-1}(V)$ is a neighborhood of x . We have $f^{-1}(V) \cap A \neq \emptyset$. Then

$$\emptyset \neq f(f^{-1}(V) \cap A) \subseteq V \cap f(A)$$

Since V is arbitrary, $f(x) \in \overline{f(A)}$. □

1.2 Homeomorphisms

Definition 1.2.1

Let X and Y be topological spaces; let $f : X \rightarrow Y$ be a bijection. If both the function f and the inverse function

$$f^{-1} : Y \rightarrow X$$

are continuous, then f is called a *homeomorphism*.

1.3 Constructing Continuous Functions

Theorem 1.3.1 ► Rules for constructing continuous functions

Let X , Y , and Z be topological spaces.

- (a) (Constant function) If $f : X \rightarrow Y$ maps all of X into the single point y_0 of Y , then f is continuous.
- (b) (Inclusion) If A is a subspace of X , the inclusion function $j : A \rightarrow X$ is continuous.
- (c) (Composites) If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then the map $g \circ f : X \rightarrow Z$ is continuous.
- (d) (Restricting the domain) If $f : X \rightarrow Y$ is continuous, and if A is a subspace of X , then the restricted function $f|_A : A \rightarrow Y$ is continuous.
- (e) (Restricting or expanding the range) Let $f : X \rightarrow Y$ be continuous. If Z is a subspace of Y containing the image set $f(X)$, then the function $g : X \rightarrow Z$ obtained by restricting the range of f is continuous. If Z is a space having Y as a subspace, then the function $h : X \rightarrow Z$ obtained by expanding the range of f is continuous.
- (f) (Local formulation of continuity) The map $f : X \rightarrow Y$ is continuous if X can be written as the union of open sets U_α such that $f|_{U_\alpha}$ is continuous for each α .